

Giorgio Maria Maioli

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Education

Eindhoven University of Technology

MSc Hons in Data Science & Artificial Intelligence

Eindhoven, NL

Sept 2025 – Exp July 2027

- **Coursework:** Generative AI Models, Deep Learning, Machine Learning Engineering, Big Data Management

Università degli Studi Roma Tre

BS in Economics and Big Data (Final grade: 109/110)

Rome, IT

Oct 2022 – March 2025

- GPA: 3.79, **graduated 6 months ahead** of schedule · Minor in Data Science
- **Erasmus+** Merit Scholarship: Mykolas Romeris University, Vilnius (Aug 2023 – Jan 2024)

Experience

AI Engineer Intern

BIP – Business Integration Partners

Rome, IT

Apr 2025 – Jul 2025

- **Architected and developed a full-stack, multi-agent platform** for dynamic document generation, **autonomously managing** the project from concept to a fully functional Proof of Concept within 3 months.
- Engineered a **novel two-stage generation pipeline**: first synthesizing source PDFs into a structured "Knowledge Exoskeleton" via **multimodal LLMs**, using it as high-quality context for document creation.
- Designed and implemented a **modular backend architecture** featuring an AgentFactory for **dynamic prompt engineering** and a ProjectManager for robust **data persistence and management**.

Research & Projects

MeditaSplit — bunq Hackathon *Agentic Bill-Splitting App*

[GitHub](#) 

- Built an end-to-end agentic bill-splitting app in under 24h: photo, **Claude Vision** receipt parsing, real Bunq payment requests, with voice input and multi-turn conversational edits.
- **Designed the full agentic loop** (8 reasoning turns, 2 tools, hard financial guardrails) and all system prompts; voice input and conversational edits supported across 3 input modes.

Subgroup Discovery with LLMs for Sentiment Analysis

[GitHub](#) 

- Developed a framework using LLMs to auto-generate features for unstructured data identifying subgroups where sentiment classification models systematically fail.

ML Engineering: Fine-Tuning CNNs, ViTs and LLMs for Medical AI

- **Fine-tuned ResNet-18** (87.1%) and **ViT-B/16** on DermaMNIST; **interpreted predictions** via Grad-CAM and Attention Rollout, exposing clinically relevant failure modes on the Melanoma/Nevi pair.
- Applied **LoRA** (via Optuna HPO) to fine-tune Gemma-3-270M-IT on MedMCQA (+8.6pp over random baseline); built a **LlamaIndex RAG pipeline** that eliminated hallucinations on knowledge-intensive medical queries.

BSc Thesis: Macrofinancial Effects of US Political Cycles via Deep Learning

- Developed and compared **TFT** and **LSTM** models on a 40-year daily dataset of economic, financial, and sentiment indicators, achieving **~6×** better performance with TFT over the LSTM baseline.
- Interpreted results via TFT native attention weights and **LIME**, surfacing regime-dependent patterns across political cycles.

Technical Skills

Programming: Python, R, SQL, MATLAB

AI & ML Frameworks: PyTorch/lightning, scikit-learn, TensorFlow, Keras, Optuna, Hugging Face, LangChain/Graph

Data Science: Pandas, NumPy, SciPy, Matplotlib, LIME, SHAP, LoRA

Database & Data Management: MySQL & Oracle SQL, ChromaDB, Spark, Hadoop

Languages: Italian (Native), English (C1)